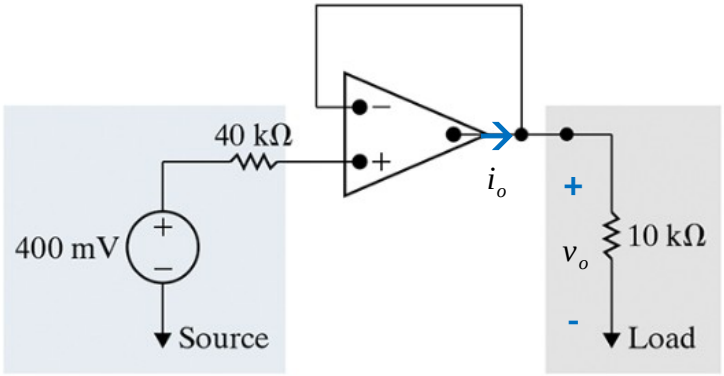
EE 2010
Fall 2026

Name Wyatt Evans

Game 23
10/2/2026

a) Find the output v_o for the circuit below. Show detailed work.
b) Find the current i_o coming out of the output pin of the op-amp. You can use KCL at the output node to determine the current.



$V_R = (40k\Omega)(0A)$
$= 0$

$V_+ = V_S - V_R = 400mV - 0V$
$= 400mV = V_- = V_o$

$gain = 1 + \frac{0}{40k\Omega} = 1$

$1 = \frac{V_o}{V_{in}}$

$V_o = V_{in}$

$\boxed{V_o = 400mV}$

$i_o = \frac{V_o}{R_L} = \frac{0.4V}{10000\Omega} = 0.00004A = \boxed{40\mu A}$